

Final Project (100 points)

Your final project consists of 3 components:

Component	Format	Submit	Points
Report	PDF	1 file via Canvas	60
Analysis Materials	Jupyter notebooks	1 or 2 notebooks via Vocareum	30
Retrospective Commentary	PDF	1 file via Canvas	10
TOTAL			100

You are not required to follow what you described in the project proposal. You can make any changes that you think will lead to a better final project. Please do not describe the changes that you made between the proposal and the final project. We will not be reviewing your proposal or the feedback we gave when grading the final project.

You may not use generative AI tools to produce any of the content that you submit in the final report. You may consult AI tools, but may not paste (or reproduce) any text, plots, or code from AI tools into your submitted work. Submissions that violate this policy will be penalized.

Report (60 points)

The first component of the assignment is your final report. This is where you describe the full scope of your project. You must upload a PDF that includes your name and your report. Where the assignment calls for you to write a paragraph, that paragraph should be 100-200 words.

Your report should include the following sections:

1. Summary

- Present a brief summary of your project. You should describe the problem that motivated the project, what you found in your analyses, and what the main impact of your analysis is. Focus on the most important details.
- Your summary should be 3-5 sentences, 100 words maximum.

2. Problem Statement

- Describe the problem that your project addresses, following the instructions from Section 1.2 of the project proposal.
- This section should be between 200-400 words. It should also cite at least 3 different sources.

3. Research Questions

- Present your 2 RQs, following the instructions from Section 2.1 of the project proposal.

4. Analysis for RQ1

- You should include:
 - 1 paragraph summarizing how you analyzed the data.
 - 1 paragraph summarizing what you found. This description should qualify your answer with any uncertainties or limitations to your analysis.
 - 2 (and only 2) visualizations that present your findings. All visualizations should be understandable and convey a clear message. For each visualization, include a 1 sentence caption. Be thoughtful and creative about how to most effectively visualize your data/analysis. (If you create a visualization with 2 plots side-by-side, that counts as 2 plots.)
 - 1 paragraph interpreting your findings. To help you get started, consider the following questions: What do the quantitative measures you found reflect in terms of real-world conditions? Do your findings align or conflict with your expectations? Do your findings inform your plans for approaching RQ2?
- Additional notes:
 - You should not present any code or raw output from python (e.g., screenshots of DataFrames). It's fine to take screenshots of your visualizations to include in the report.
 - Do not include any tables.
 - Your analyses don't need to present statistically significant results. Non-significant results are important to note, especially if they run counter to expectations. Don't hack your analysis just to arrive at a significant statistic. You don't want to make recommendations that aren't supported by the data.

5. Analysis for RQ2

- Follow the same instructions as RQ1.

6. Conclusion: Recommendations

- Conclude your report with 2 recommendations. This is your chance to highlight the key impacts of your analysis.
- For each recommendation:
 - Provide a clear header that presents each recommendation (e.g., "Recommendation 1: XXX").
 - Clarify who the audience is for your recommendation. This should be a person or group of people who are equipped to act on your recommendation (e.g., "Audience: XXX").
 - Provide 1 paragraph justifying and explaining each recommendation. You should make it clear how the recommendation follows from your analyses. Be specific about how your audience should act on your recommendation.

Analysis Materials (30 points)

The second component of the assignment is your analysis. You must complete your analysis using Jupyter notebooks in Vocareum. You can present your analysis all in one notebook or in multiple notebooks. (Having one notebook for each RQ makes it easier to read. However, if you are using the same data across multiple RQs, it will be easier to keep the full analysis in one notebook.) Either way, you should use Markdown headers to make it clear what you are working on at each part of your notebooks.

You should present a complete analysis for all of your RQs. Your analysis should include any applicable robustness checks (e.g., considering alternative explanations for your findings, exploring whether different levels of aggregation affect your results, and exploring whether different ways of defining metrics affect your results).

You should present the code you use to generate the visualizations that you present in your final report. However, you will not be graded on the quality of the visualizations in your notebooks. There is no limit to the number of visualizations that can be included in your notebooks.

Your code should be well-organized and should include comments explaining what each code block is for. You should explain your reasoning and justification for the steps you take. This is particularly important when you are making judgment calls such as dropping observations or defining metrics.

You are not required to provide any Markdown cells that explain/interpret your results in these Jupyter notebooks. Your discussion of the results goes into your final report.

(NOTE: Students will lose access to Vocareum at the end of the semester. We recommend that you download all of your notebooks at the end of the semester so that you can access those files in the future. We will not be able to retrieve them once you lose access.)

Retrospective Commentary (10 points)

The third component of the assignment is a short reflection on lessons learned during the project. You must describe 2 lessons. For each lesson, explain why you learned that lesson and how the lesson will inform your work in the future.

These lessons should focus on the process of working on your project, not the substance of your findings. For instance, you should not describe learning about labor unions as a lesson.

To help you identify lessons, reflect on the following questions:

- What challenges did you encounter?
- What parts of the project were harder than you expected?
- What parts of the project took more time than you expected?
- What would you do differently the next time you work on a similar project?

You must upload a PDF that includes your name and your reflection on Canvas. Your commentary should be between 150-250 words total.

NOTE on commentary: We are interested in understanding how you learned the lessons described, and how they will apply those lessons in the future. This is about the learning process and how you identify takeaways from a project to improve your work in the future. We aren't looking for "correct" lessons learned, but instead on how you communicate your learning process.

Sharing the report with others

The course staff would like to share some final reports with other groups, such as the Engaged Learning Office and our project partners outside of UM. These groups are interested in reviewing final projects and learning from your analyses.

If you would prefer that your project NOT be shared with anyone, please indicate that clearly near the top of your final report. This decision will not affect your grade. If you do not specify, we will assume you are comfortable with us sharing your report.

Grading Rubric

Report (60 points total)

Formatting (2 points)

Excellent (2 points)	There are clear headers for each section of the report. It's easy to tell what section we're in at all times. The document is pleasant to read.
Acceptable (1 points)	It's difficult to follow along in the report. There are not enough headers, and/or there are several places where it's hard to tell what section we're in. The document is mostly pleasant to read, but has some awkward formatting issues (e.g., text randomly changes font or size with no explanation).
Unacceptable (0 points)	It takes a lot of work to follow along in the report. There are minimal or no headers that delineate each section from one another.

Summary (3 points)

Excellent (3 points)	Clearly and concisely summarizes the problematic situation, main findings, and impact within the word limit.
Good (2 points)	Summarizes the key points but may miss one minor detail or slightly exceed the word limit.
Acceptable (1 points)	Provides a summary but lacks clarity or misses key details. May exceed the word limit by a significant margin.
Unacceptable (0 points)	Incomplete or unclear summary, missing several key points, or not within the word limit. If the problem does not connect to the theme topic, you will receive 0 points for this section.

Problem Statement (10 points)

Excellent (10 points)	Clearly describes the problem, cites at least 3 sources, and is within the word limit.
Good (9 points)	Describes the problem well but lacks a minor detail, or only cites 2 sources, or slightly off the word limit.
Acceptable (7-8 points)	Problem statement is vague or lacks clarity, cites 1 source or none, or significantly off the word limit.
Unacceptable (0-6 points)	Incomplete problem statement, lacks sources, or far exceeds or falls short of the word limit.

Research Questions (5 points)

Excellent (5 points)	Both research questions are clear and directly relevant to the problem statement.
Good (4 points)	The research questions are presented and mostly relevant, but there might be minor issues in clarity or relevance.
Acceptable (3 points)	Research questions are somewhat clear but have noticeable clarity or relevance issues.
Unacceptable (0-2 points)	Research questions are unclear, irrelevant, or missing.

Analysis (30 points total; 15 points for each RQ)

Excellent (15 points)	Description of analysis and findings is clear and comprehensive. It provides convincing answers to the RQs. There are 2 effective visualizations with captions, and a well-written interpretation.
Good (13-14 points)	Description of analysis and findings is mostly clear, comprehensive, and convincing. There are 2 visualizations with captions, though minor details are lacking or slightly unclear.
Acceptable (11-12 points)	The analysis and findings are summarized but lack clarity. Visualizations are present but may not effectively convey the message, and the interpretation is vague.
Unacceptable (0-10 points)	Incomplete analysis, unclear findings, poor or missing visualizations, or lack of interpretation.

Conclusion & Recommendations (10 points)

Excellent (10 points)	Two clear, justified recommendations with specified audiences and well-explained actions.
Good (9 points)	Recommendations are clear but may lack a minor detail in justifying or explaining actions.
Acceptable (7-8 points)	Recommendations are present but lack clarity, audience specification, or detailed justification.
Unacceptable (0-6 points)	Incomplete recommendations, unclear or missing justifications, or undefined audience.

Analysis Materials (30 points)

Excellent (29-30 points)

The analysis provides a convincing answer to the RQs. Code is well-organized, clean, and comprehensively commented. The analysis is thorough, with relevant robustness checks and clear headers.

Good (26-28 points)

The analysis mostly answers the RQs, but has some errors or leaves some open questions. Code is well-organized but may miss some comments or robustness checks. The analysis is mostly thorough with a few minor gaps.

Acceptable (23-25 points)

Basic organization but lacks sufficient comments or robustness checks. The analysis may be somewhat incomplete.

Unacceptable (0-22 points)

Poorly organized code with little to no comments or robustness checks. The analysis is incomplete or poorly executed.

Retrospective Commentary (10 points)

Excellent (10 points)

Provides 2 insightful lessons learned, with clear explanations, well-connected to the project, and strongly indicative of future improvements. Adheres to the word limit.

Good (9 points)

Provides 2 lessons learned that are adequately explained but may lack depth or future applicability.

Acceptable (7-8 points)

Provides 1-2 lessons learned, with basic explanations and unclear connection to future work. May not adhere to word limit.

Unacceptable (0-6 points)

Vague or missing lessons, poor explanations, and little to no connection to future improvements.